B.A. -Mathematics-Session 2023-24
Syllabus For Ist Sem (As per NEP-2020)

CG-M MA 1 : CALCULUS – I

Credits = 4 (L = 3 + T = 1)

1 Credit of Lecture = 1 hr. and 1 Credit of Tutorial = 3 hrs.

So, 4 Credits = 3 hrs. of lectures and 3 hrs. of Tutorials in a week = 6 hrs. /week

Total Contact hours in a semester of 15 weeks = 90 hrs. (45 hrs. Lectures + 45 hrs. Tutorials)

Exam Time: 3 hours **Max. Marks: 100**

(90 External + 10 internal assessment)

Note: 1. The syllabus has been split into four Units: Unit – I, Unit – II, Unit - III and Unit - IV. The examiner will set in total nine questions including a compulsory question. The compulsory question (Question No.1) will contain 6 parts covering the entire syllabus. Two questions will be set from each Unit.

2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.

Objective: The main goal of this course is to deliver basics of Limits, Continuity and Differentiability of functions in one variable as well as in two variables. The course also covers the basic concepts of curve tracing and vector calculus.

Learning Outcomes:

This course will enable students to understand the concepts of Limits and continuity. Students will be able to understand the applications of differential and vector calculus. The course will enable students to determine derivatives of two variable functions, extreme values and understand the concepts of tangents, normals, asymptotes and curve tracing.

UNIT-I

Order properties of Real Numbers, least upper bound, greatest lower bound, order completeness property of real numbers, Archimedean property of Real Numbers.

ε-δ definition of limit, one-sided limit, limits involving infinity, continuity, types of discontinuities, define derivative of a function, tangents and normals at a point.

UNIT-II

Rolle's Theorem, Lagrange's Mean Value Theorem and Cauchy's Mean Value Theorem. Their geometric interpretation and applications.

Convexity and concavity, points of inflexion, asymptotes, tracing of curves in Cartesian coordinates. Polar Coordinates, Curve tracing in Polar coordinates.

UNIT-III

Limits and continuity for functions of two variables, partial derivatives, the chain rule, extreme values and saddle points of functions.

UNIT-IV

Vector differentiation, gradient, divergence, curl, directional derivative, tangent planes, properties and applications.

SUGGESTED READINGS

1. G. B.THOMAS, M. D. WEIR and J. R. HASS, THOMAS, *Thomas' Calculus*, (12th Edition), Pearson, 2014.
2. J. D. MURRAY and M. R. SPIEGEL, *Theory and Problems of Advanced Calculus*, Schaum's Outline Series, Schaum Publishing Co., New York.
3. S. NARAYAN and P.K. MITTAL, *Differential Calculus*, Ed 2006, S. Chand & Co., New Delhi.
4. M. J. STRAUSS, G. L. BRADLEY and K. J. SMITH, *Calculus*, (3rd Edition), Pearson Education, 2007.

B.A. -Mathematics-Session 2023-24

Syllabus For IInd Sem (As per NEP-2020)

CG-M MA 2 : ALGEBRA

Credits = 4 (L = 3 + T = 1)

1 Credit of Lecture = 1 hr. and 1 Credit of Tutorial = 3 hrs.

So, 4 Credits = 3 hrs. of lectures and 3 hrs. of Tutorials in a week = 6 hrs. /week

Total Contact hours in a semester of 15 weeks = 90 hrs. (45 hrs. Lectures + 45 hrs. Tutorials)

Exam Time: 3 hours **Max. Marks: 100**

(90 External + 10 internal assessment)

Note: 1. The syllabus has been split into four Units: Unit – I, Unit – II, Unit - III and Unit - IV. The examiner will set in total nine questions including a compulsory question. The compulsory question (Question No.1) will contain 6 parts covering the entire syllabus. Two questions will be set from each Unit.

2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.

Objective: The course introduces the basics of linear algebra. It aims at understanding the multiplicity of roots and introduces the basic tools of solving cubic and bi-quadratic equations. The course introduces the concept of Eigen values and Eigen vectors of a matrix. The objective is to understand nature of solution of a linear system of equations.

Learning Outcomes:

This course will enable students to identify multiple roots and solve cubic and bi-quadratic equations. The students will be able to determine the rank of a matrix, Eigen values, Eigen vectors and solve linear system of equations using matrices.

UNIT-I

Euclid's algorithm, synthetic division, roots and their multiplicity. Complex roots of real polynomials occur in conjugate pairs with same multiplicity. Relation between roots and coefficients. Transformation of equations.

UNIT-II

Solution of cubic and bi-quadratic equations, Cardan's method of solving a cubic, trigonometric solutions of a real cubic with real roots, discriminant and nature of roots of a real cubic. Descarte's and Ferrari's method for solving a bi-quadratic.

UNIT-III

Hermitian and Skew-Hermitian matrices, linear dependence of row and column vectors, row rank, column rank and rank of a matrix and their equivalence. Theorems on consistency of a system of linear equations (both homogeneous and non-homogeneous).

UNIT-IV

Eigen-values, Eigen-vectors and characteristic equation of a matrix, Cayley-Hamilton theorem and its use in finding inverse of a matrix. Diagonalization.

SUGGESTED READINGS

1. S.R. KNIGHT and H.S.HALL, *Higher Algebra*, H.M. Publications,1994.
2. S.NARAYAN and P.K.MITTAL, *A Text Book of Matrices*, Ed2007, S.Chand & Co., New Delhi.
3. CHANDRIKA PRASAD, *Text Book on Algebra and Theory of Equations*, Pothishala Private Ltd. Allahabad.
4. V.SAHAI and V.BIST, *Linear Algebra*, 2nd Edition, Narosa Publishing House, 2013.
5. A.KUROSH, *Higher Algebra*, MIR Moscow, 1982.

CG-M MDC-1 : BASIC MATHEMATICS – I

BA SEMESTER – Ist

Credits = 3 (L = 2 + T = 1)

1 Credit of Lecture = 1 hr. and 1 Credit of Tutorial = 3 hrs.

So, 3 Credits = 2 hrs. of lectures and 3 hrs. of Tutorials in a week = 5 hrs. /week

Total Contact hours in a semester of 15 weeks = 75 hrs. (30 hrs. Lectures + 45 hrs. Tutorials)

Exam Time: 3 hours **Max. Marks: 75**

(68 External + 7 internal assessment)

Note:
1. The syllabus has been split into three Units: Unit – I, Unit – II and Unit - III. The examiner will set in total seven questions including a compulsory question. The compulsory question (Question No.1) will contain 7 parts of 2 marks each, covering the entire syllabus. Two questions, each of 18 marks, will be set from each of the 3 Units.
2. A student will be required to attempt four questions in all, selecting one question from each Unit and the compulsory question.

Objective : The objective of this course is to study the basics of various topics of Mathematics which is foundation for further learning in mathematics, physics, statistics etc

Learning outcomes:

The course will enable the students to learn the following:

1. Application of set theory and permutation combination.
2. Basics of matrices and its determinant.
3. Basic statistics, mean deviation, variance and standard deviation.
4. Graphical representation of solution of linear inequalities and its application.

Unit –I

Sets, Relations and functions, Principal of Mathematical Induction, Permutation and Combinations, {Scope as in chapters 1, 4, 7 of (A)}. Matrix, type of Matrices, operations on matrices, Transpose of Matrix, Symmetric and skew symmetric Matrices. {Scope as in chapters 3 (3.1 to 3.6) of (B) Part-I}

Unit –II

Determinant, Properties of Determinants, Minor and Cofactors, Adjoint and Inverse of a Matrix {Scope as in Chapter 4 (4.1 to 4.6) of (B) Part-I}. Introduction of Statistics, Measure of Dispersion, Range, Mean Deviation, variance and Standard Deviation, Analysis of Frequency Distribution.

Unit – III

Linear Inequalities – Introduction, Algebraic Solution of Linear Inequalities in one variable and their Graphical Representation, Graphical Solution of Linear Inequalities in two

variables, Solution of System of Linear Inequalities in two variables {Scope chapters 6}

Essential Text books

1. Mathematics; A Text book for class XI, NCERT, 2006
2. Mathematics; A Text book for class XII, NCERT (Part I and Part II), 2006

CG-M MDC-2 : BASIC MATHEMATICS – II

BA SEMESTER – IInd

Credits = 3 (L = 2 + T = 1)

1 Credit of Lecture = 1 hr. and 1 Credit of Tutorial = 3 hrs.

So, 3 Credits = 2 hrs. of lectures and 3 hrs. of Tutorials in a week = 5 hrs. /week

Total Contact hours in a semester of 15 weeks = 75 hrs. (30 hrs. Lectures + 45 hrs. Tutorials)

Exam Time: 3 hours **Max. Marks: 75**

(68 External + 7 internal assessment)

Note:
1. The syllabus has been split into three Units: Unit – I, Unit – II and Unit - III. The examiner will set in total seven questions including a compulsory question. The compulsory question (Question No.1) will contain 7 parts of 2 marks each, covering the entire syllabus. Two questions, each of 18 marks, will be set from each of the 3 Units.
2. A student will be required to attempt four questions in all, selecting one question from each Unit and the compulsory question.

Objective: This course is designed to introduce the fundamental concepts of limits continuity, differentiation, integration of functions of one variable and Probability. Its objective is to acquaint students with various applications of these topics relating to extreme value problems, problems related to Probability.

Learning outcomes :

The course will enable the students to learn the following :-

1. Idea of limit continuity derivative and application of derivatives (increasing and decreasing of function, maximum, minimum, Rolle's Theorem).
2. Integration as anti derivative and process of integration.
3. Basic Idea of probability.

Unit-I

Limits, continuity, Derivative and differentiability (Excluding Trigonometric Functions). Introduction, intuitive Idea of derivative, Limits, Continuity, Differentiability. Exponential and Logarithmic Differentiation, Derivative of a function in parameter, Second order derivative, (Scope as in Chapter 13 of (A) & Chapter 5 Part-I of (B).

Unit-II

Application of derivatives: Increasing and decreasing functions, Maxima and Minima, Rolle's Theorem and Lagrange's Mean Value Theorem (without proofs only geometric interpretation and problems). (Scope as in Chapter 6 of (B) Part-I)

Unit-III

Integral Calculus –(Excluding Trigonometric Functions). Integral as anti- derivative,

Integration by substitution, by partial fractions and by parts. (Scope as in Chapter 7 & 8 of Part II of (B)).

Probability – Introduction, Random Experiments, Event, Axiomatic Approach to Probability (Scope as in Chapter 16 of (A)).

Essential Textbooks

1. Mathematics , A Text book for class XI , NCERT ,2006
2. Mathematics , A Text book for class XII , NCERT (Part I and Part II) ,2006

CG-M : SEC-1:DISCRETE MATHEMATICS
BA SEMESTER – Ist

Credits = 3 (L = 2 + T = 1)

1 Credit of Lecture = 1 hr. and 1 Credit of Tutorial = 3 hrs.

So, 3 Credits = 2 hrs. of lectures and 3 hrs. of Tutorials in a week = 5 hrs. /week

Total Contact hours in a semester of 15 weeks = 75 hrs. (30 hrs. Lectures + 45 hrs. Tutorials)

Exam Time: 3 hours **Max. Marks: 75**

(68 External + 7 internal assessment)

Note:
1. The syllabus has been split into three Units: Unit – I, Unit – II and Unit - III. The examiner will set in total seven questions including a compulsory question. The compulsory question (Question No.1) will contain 7 parts of 2 marks each, covering the entire syllabus. Two questions, each of 18 marks, will be set from each of the 3 Units.
2. A student will be required to attempt four questions in all, selecting one question from each Unit and the compulsory question.

Objective: The objective of this course is to acquaint the students with the basic concepts in Discrete Mathematics and Graph Theory. The course discusses some important applications of graph theory.

Learning Outcomes:

The course will enable students to understand sets, relations and functions. The students will be able to understand the basics of graph theory.

Unit-I

Sets, properties of relations: injective, surjective and bijective functions, compositions. Recurrence Relations and Recursive Algorithms - Linear-Recurrence Relations with Constant Coefficients; Homogeneous Solutions: Particular Solution, Total Solution and Solution by the Method of Generating functions.

Unit-II

Graphs – Basic Terminology, Multi-graphs, Weighted Graphs, Paths and Circuits, Shortest Paths, Eulerian Paths and Circuits. Travelling Salesman Problem.

Unit-III

Planar graphs, Spanning Trees, Prim's Algorithm for generating minimum weights panning graphs, Digraphs and Chromatic numbers.

SUGGESTED READINGS

1. A. Doerr and L. Kenneth, Applied Discrete Structures for Computer Science, 1989, Galgotia Publications Pvt. Ltd.
2. C. L. Liu, 1985, Elements of Discrete Mathematics, McGraw Hill.
3. S. Lipschutz and Lipson: 2000 Solved Problems in Discrete Mathematics, McGraw-Hill, 1992.

CG-M : SEC-2 : MATHEMATICAL MODELING
BA SEMESTER – IInd

Credits = 3 (L = 2 + T = 1)

1 Credit of Lecture = 1 hr. and 1 Credit of Tutorial = 3 hrs.

So, 3 Credits = 2 hrs. of lectures and 3 hrs. of Tutorials in a week = 5 hrs. /week

Total Contact hours in a semester of 15 weeks = 75 hrs. (30 hrs. Lectures + 45 hrs. Tutorials)

Exam Time: 3 hours **Max. Marks: 75**

(68 External + 7 internal assessment)

Note:
1. The syllabus has been split into three Units: Unit – I, Unit – II and Unit - III. The examiner will set in total seven questions including a compulsory question. The compulsory question (Question No.1) will contain 7 parts of 2 marks each, covering the entire syllabus. Two questions, each of 18 marks, will be set from each of the 3 Units.
2. A student will be required to attempt four questions in all, selecting one question from each Unit and the compulsory question.

Objective : The course content is designed to introduce the concept of mathematical modelling. The course explains fundamental techniques of solving basic differential equations and different mathematical models.

Learning Outcomes:

Students will be able to solve first order ordinary differential and initial value problems. Students will be able to learn the basic techniques of Mathematical Modeling and also its limitations. Students will be able to learn basic population and epidemic models. Students will be able to learn the concepts of equilibrium points and stability and their importance in Mathematical Modeling.

UNIT-I

First order ordinary differential equations, Geometric Interpretation, Separable differential equations, Exact differential equations, Initial Value Problems (IVP).

Simple situations requiring mathematical modeling, The technique of mathematical modeling, Classification of mathematical models, Limitations of mathematical modeling.

UNIT-II

Mathematical modeling through ordinary differential equations, Mathematical modeling in population dynamics, Growth and decay models. Equilibrium points and stability, Limited growth model with harvesting, Discrete population growth and chaos, Time delay regulation. (Section 2.11, 3.3, 3.6, 3.7 of [1])

UNIT-III

Prey-predator models, Competition models, Mathematical modeling of epidemics through system of Ordinary Differential Equations, A simple Epidemic model, SIS and SIR models, Epidemic model of influenza and its application. (Section 5.1, 5.2, 5.3, 5.5 of [1])

SUGGESTED READINGS

1. B. Barnes & G. R. Fulford: Mathematical modelling with case studies (using Maple & Matlab) CRC press, Taylor and Francis, 2009.

2. S.L. Ross, Differential Equations, 3rd Ed., John Wiley and Sons, India, 2004.

3. E. A. Coddington, An introduction to ordinary differential equation, Prentice- Hall of India.

4. J. N. Kapur, Mathematical modeling, New age international publishers, New Delhi, 2013.

5. E. Kreyszig ,*Advanced Engineering Mathematics*, John Wiley & Sons Inc., New York, 1999.

6. D.A. Murray, *Introductory Course on Differential Equations*, Orient Longmen, (India), 1967.